

An Anatomy- and Topology-Preserving Framework for Coronary Artery Segmentation

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Abstract—Coronary artery segmentation is critical for coronary artery disease diagnosis but challenging due to its tortuous course with numerous small branches and inter-subject variations. Most existing studies ignore important anatomical information and vascular topologies, leading to less desirable segmentation performance that usually cannot satisfy clinical demands. To deal with these challenges, in this paper we propose an anatomy- and topology-preserving two-stage framework for coronary artery segmentation. The proposed framework consists of an anatomical dependency encoding (ADE) module and a hierarchical topology learning (HTL) module for coarse-to-fine segmentation, respectively. Specifically, the ADE module segments four heart chambers and aorta, and thus five distance field maps are obtained to encode distance between chamber surfaces and coarsely segmented coronary artery. Meanwhile, ADE also performs

coronary artery detection to crop region-of-interest and eliminate foreground-background imbalance. The follow-up HTL module performs fine segmentation by exploiting three hierarchical vascular topologies, *i.e.*, key points, centerlines, and neighbor connectivity using a multi-task learning scheme. In addition, we adopt a bottom-up attention interaction (BAI) module to integrate the feature representations extracted across hierarchical topologies. Extensive experiments on public and in-house datasets show that the proposed framework achieves state-of-the-art performance for coronary artery segmentation.

Index Terms—Coronary artery segmentation, hierarchical topology representation, anatomical dependency encoding, multi-task learning, coarse-to-fine segmentation.

I. INTRODUCTION

CORONARY artery disease (CAD) is one of the most critical cardiovascular diseases in our daily life, which can lead to myocardial ischemia, hypoxia, or necrosis in severe cases [1], [2]. To evaluate the severity of CAD, coronary computed tomography angiography (CCTA) as a non-invasive imaging technique is popularly adopted to reconstruct and quantitatively assess the coronary artery structures of patients [3], [4]. Coronary artery segmentation is a crucial prerequisite step for CAD analysis since it enables diameter measurement of vascular lumen [5], [6], vascular branch labeling [7], and centerline tracing [8] for construction of the coronary tree, delineation of the vascular skeleton topology, and quantification of the coronary stenosis grading, respectively. Therefore, coronary artery segmentation plays an essential role for the diagnosis and treatment of CAD.

To date, automatic coronary artery segmentation from CCTA images remains challenging mainly due to three reasons as shown in Fig. 1. **Challenge 1: Small volume ratio.** In CCTA images, the foreground volume occupied by coronary arteries is very much small compared to the whole image (less than 0.05%), which makes the background and the vascular region extremely imbalanced, so that the deep learning-based methods are usually prone to bias and result in inferior segmentation performance for the foreground vascular regions. **Challenge 2: Thin structures.** Coronary arteries have tortuous dendritic tubular structures with diverse variations in amount, length, size, and elongated lumen, usually less than 5.0mm in diameter (Fig. 1(b)). These special structures induce difficulty to learn the context information along the

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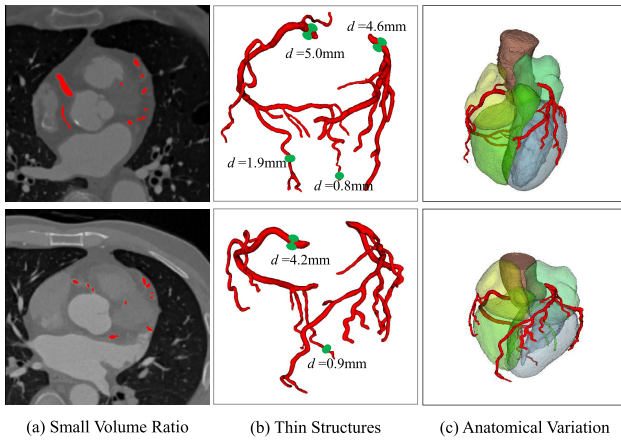


Fig. 1. The challenges of accurate coronary artery segmentation. (a) Small volume ratio: the ratio between vascular region and background is heavily imbalanced; (b) Thin structures: vascular lumen diameters ranging from 0.8 mm to 5 mm; (c) Inter-subject anatomical variation: large divergence of vessel in shape and location across individuals.

vessel direction, which leads to numerous hard-to-segment regions and discontinuous segmentation in practice, especially at the small branches of coronary artery. **Challenge 3: Inter-subject anatomical variation.** Coronary arteries originate from aorta, and are usually divided into left and right coronary arteries. They are intertwined with or attached to multiple heart chambers (Fig. 1(c)), *i.e.*, atria, and ventricles. In fact, the anatomical structures of coronary artery such as shape, abundance, and path are patient-specific with large inter-subject variability, which result in significant diversity in the spatial distribution of vessel and make coronary artery segmentation more challenging. Therefore, designing an accurate coronary artery segmentation framework accounting for the above difficulties is an urgent and essential task for subsequent analysis and diagnosis.

Many traditional methods for coronary artery segmentation have been proposed in the past, *e.g.*, based on filter enhancement [9], region-growing [10], and centerline guidance [11]. Although these approaches have achieved reasonable performance in coronary artery segmentation, their performance is limited by hand-crafted features based on prior knowledge and parameter tuning. Most deep learning-based methods for medical image segmentation [12] are built on UNet and its variants [13]. These models perform well for general segmentation tasks but not for coronary artery segmentation due to the aforementioned task-specific difficulties.

With current advances in deep learning, some existing studies attempted to tackle these difficulties via customized networks [14]. For example, some attention-based methods [15], [16], [17] were proposed to learn more fine-grained feature representations in coronary artery segmentation. However, these methods ignored important coronary anatomical and topological properties, resulting in limited performance improvement. Some other studies employed topology representations [18], [19] as additional loss constraint, which intends to deal with the discontinuity of the segmented vascular lumen. However, the topological constraints applied in these methods are incomplete and cannot fully describe the overall topology of the coronary arteries, which makes it

difficult to ensure the integrity and continuity of the segmentation results.

To address the aforementioned challenges, in our previous work [20], we developed a coronary artery segmentation framework that consists of an anatomical dependency encoding (ADE) module and a hierarchical topology learning (HTL) module for coarse-to-fine segmentation. The ADE module performs coarse segmentation and generates distance field maps between heart chambers and segmented vessel, and the HTL enhances segmentation by learning hierarchical topologies, including key points, centerlines, and connectivity. However, there are two main limitations with our previous work: 1). Although the bifurcation points and endpoints (denoted as key points), the centerlines, and the connectivity with neighbors represent hierarchical tree-like topology, skeleton structure, and vessel continuity, respectively, these complementary representations are not interactively integrated with each other, which cannot exploit multi-level topological morphology; 2). Limited by the model design, the number of key points has to be predefined, which causes less flexibility. Herein, we extend our previous work to address the above limitations. Specifically, we propose a bottom-up attention interaction (BAI) module within the HTL module to enhance the intrinsic connections between different levels of topological representations. In addition, we introduce the key-point detection method via weighted Hausdorff distance loss within the HTL module, which bypasses the requirement of a predefined number of key points. Moreover, we verify the effectiveness of the proposed framework on two large datasets and perform in-depth analysis and ablation study quantitatively and qualitatively. The main contributions of this work can be summarized as follows:

- An anatomy- and topology-preserving framework for coronary artery segmentation is proposed. It consists of an anatomical dependency encoding (ADE) module and a subsequent hierarchical topology learning (HTL) module following a coarse-to-fine segmentation scheme.
- The elaborately designed ADE module performs *not only* a coarse coronary artery segmentation within a cropped region-of-interest, *but also* generates five distance field maps encoding anatomical dependency between coronary artery and segmented aorta and heart chambers.
- The follow-up HTL module introduces hierarchical topology constraints including key points, centerlines, and neighbor connectivity under a multi-task learning scheme. Besides, a bottom-up attention interaction (BAI) module is utilized to integrate these hierarchical topological representations; a novel key-point detection method is introduced to precisely locate the key point positions regardless of the number of these points.
- Extensive experiments show that our framework outperforms the state-of-the-art methods significantly.

II. RELATED WORK

In the past two decades, many methods have been proposed for coronary artery segmentation [14], [21], [22], [23]. Compared with traditional methods, deep learning-based methods have become the favorite of current research due to their

outstanding performance. These deep learning-based methods can be mainly split into two categories: 1) Attention-based methods, which intend to facilitate feature extraction based on attention mechanism; 2) Topology-based methods, which exploit the vascular topological structures by mining the morphological characteristics as additional constraints.

A. Attention-Based Deep Learning Methods

Attention mechanism enhances the representation ability of networks by emphasizing the region-specific information and ignoring the irrelevant regions [24], and its efficacy has been proven in many medical image analysis tasks [25], [26]. Li et al. [27] proposed a self-attention encoder module and sliced mosaic permutation strategy to alleviate the problem of category imbalance. However, it has difficulty in coping with discontinuity issues especially in small vessel branches. In [28], Zhu et al. developed a pyramid module to extract multi-level detailed feature representation, but overlooking the extreme imbalance between the background and the foreground vascular region. Similarly, Mou et al. [29] proposed a specific channel and spatial attention to integrate vascular local features with their global dependency. Although the attention mechanism is applied to improve the inter-class discrimination and intra-class aggregation abilities of vessel, the lack of topological continuity constraints limits further improvement of segmentation performance. In [30], Song et al. proposed a two-stage method consisting of a classification and a follow-up segmentation network, in which the vascular slices of interest are selected by a specific classifier to reduce the background interference, and then a rectifier UNet is introduced to achieve coronary artery segmentation. Due to the lack of topology-preserving constraints, it still needs to be improved in the segmentation continuity of thin vessel.

The above attention-based methods employ complex network structures (*e.g.*, attention and multi-scale mechanism) to mine local-global and multi-level feature representation in the image, which partially alleviates the issue of imbalanced vessel-to-background voxel ratio. However, these methods often overlook the anatomical and morphological features of vessel, resulting in suboptimal continuity of segmented vessel and limiting their segmentation performance.

B. Topology-Based Deep Learning Methods

Inspired by the pioneering works [31], [32], [33], more recent works attempted to learn anatomical and morphological characteristics of the vessel from topological structure representations. Kong et al. [34] employed a pre-trained UNet to extract vascular centerlines, which were then used to construct the graph-based structural connections of vessel. Subsequently, a specially designed tree-structured ConvGRU module was employed for vessel segmentation. It turns out that the quality of centerline extraction plays a crucial role for segmentation performance. In [35], Li et al. proposed a deep 3D vessel segmentation network based on bidirectional convolutional long short-term memory. It improves the accuracy of vessel segmentation by mining potential context information in both the segmentation and boundary prediction

branches. In TopNet [36], a multi-task deep learning architecture employs shared encoders and task-specific decoders for vessel segmentation, centerline extraction, and topology distance constraint. However, the lack of exploration into additional topological representations restricts its potential for achieving further performance improvement. Shit et al. [37] proposed a connectivity-aware similarity measure constraint, which is defined based on vessel centerline and formulated as a differentiable soft-cIDice. However, the proposed method uses only the skeleton constraint but not hierarchical topological information, which limits their performance. Topology-based deep learning methods can mitigate over-segmentation and under-segmentation by preserving the topology of complex tree-structured vessel.

The previous topology-based methods only consider simple topological properties of vessel, such as the most commonly employed vessel centerline, and boundary, while other types of topological elements, *e.g.*, bifurcation points, endpoints, and connectivity, have not received enough attention. As single topological property may not provide sufficient constraint for vascular segmentation, it is desirable to exploit structured topological features and combine them into hierarchical representations of the vascular topology. In such a way, neural networks can obtain enriched and robust features and further push the performance limit of coronary artery segmentation.

III. METHODS

In order to deal with the challenges as mentioned in Section I and overcome the limitations of existing methods as described in Section II, we propose a coarse-to-fine segmentation framework considering contextual anatomical dependency and hierarchical topological constraints. Specifically, the proposed segmentation framework consists of three modules as shown in Fig. 2: 1) anatomical dependency encoding module (ADE), 2) hierarchical topology learning module (HTL), and 3) bottom-up attention interaction module (BAI). The ADE module predicts the heart mask and coarse artery mask to encode the anatomical dependency into distance field maps and meanwhile obtains the coarse-grained spatial localization of vessel, thus removing irrelevant background. The output of ADE is fed into HTL to refine the coarse segmentation by learning hierarchical topological representations, covering key points, centerlines, and neighbor connectivity. The BAI module is introduced to integrate the three-level topological features within HTL.

A. Anatomical Dependency Encoding

Coronary artery usually accounts for less than 0.05% of the whole volume as demonstrated in Fig. 1(a). A preliminary coarse segmentation can locate the coronary artery and a cropping of the coarse segmentation can hence reduce the background interference. In addition, coronary artery wraps around and attaches to the surfaces of atria and ventricles, as shown in Fig. 1(c). The artery-to-heart anatomical dependency conveys critical contextual and spatial information to help locate vessel. Herein, we propose anatomical distance field maps based on such dependency. Therefore, the proposed

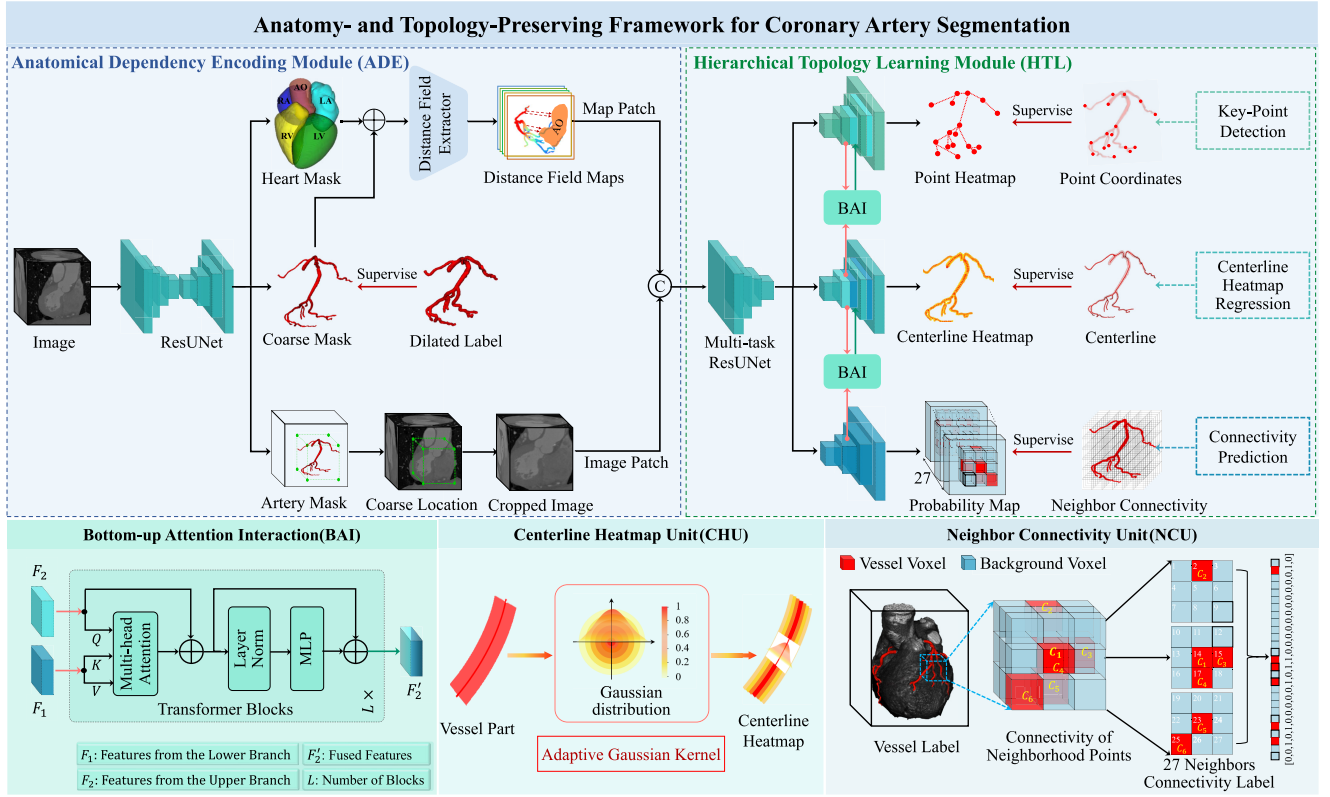


Fig. 2. Overview of the proposed framework. It consists of three components: 1) ADE module for coarse-grained localization of coronary artery and distance field map estimation; 2) HTL module for hierarchical topology-preserving including key-point detection, centerline regression, and neighbor connectivity prediction; 3) BAI module for integration and propagation of the topological representations from the enriched bottom branch to the sparse top branch of HTL.

ADE module *not only* generates distance field maps, *but also* locates the region-of-interest based on the coarse segmentation.

1) Coarse Location: We employ two separate ResUNets [13], one for heart chamber mask prediction and another one for coarse coronary artery mask prediction. Inspired by [38], we dilate the ground-truth mask (denoted as “Dilated Label” in Fig. 2) to expand the vessel segmentation area for ensuring the inclusion of all vessel voxels, especially the fine vessel branches. Specifically, The Dilated Label is generated by dilating the ground truth with a dilated kernel of size $5 \times 5 \times 5$ to supervise the coarse coronary artery segmentation. The coarse vessel segmentation can achieve a mean Dice score of 87.36% under the supervision with dilated label, and obtain more than 92.86% of the dilated coronary artery regions in the original data as shown in Fig. 5. For segmentation of heart chambers, we obtain a mean Dice score of 95%, which is empirically sufficient for subsequent fine-grained segmentation. Besides, in order to make the cropped patches contain complete coronary artery and required contextual information, we use the union of the outer rectangular boxes of the coarse vessel mask and the heart mask to crop the original image, which usually reduces the background interference by up to 50.66%.

2) Anatomical Distance Field Map: Given a coarse vessel segmentation from “Coarse Location”, we can extract a point set of coronary artery $P = [p_1, p_2, \dots, p_N]$ with a dimension

of $N \times 3$, where $N \in [26000, 140000]$. Then, we extract five point sets from the surface of each of the segmented heart chambers and aorta, *i.e.*, left ventricle, right ventricle, left atrium, right atrium, and the whole aorta, denoted as $\{O_1, O_2, \dots, O_5\}$. Each point set of heart chamber surfaces is subsampled to 8000 points using the farthest point sampling, which reduces the computational cost while preserving the essential information. The distance field maps are obtained by calculating the minimum distance between each point p_n and all the points in the individual chamber point set $\{O_1, O_2, \dots, O_5\}$ respectively as defined by Eq.(1) such that we get five maps in the end. These maps are normalized and then clipped by an upper bound of 1 to reduce the interference of outliers as formulated in Eq.(2) and then concatenated together to form five distance field maps F_a , denoted as $F_a = \{d_{n,1}, d_{n,2}, \dots, d_{n,5}\}$. The calculation of such a distance field map between P and individual chamber O_I ($I = 1, 2, \dots, 5$) is formulated as

$$S_{n,I} = \min_{o_i \in O_I} \sqrt{(p_n^x - o_i^x)^2 + (p_n^y - o_i^y)^2 + (p_n^z - o_i^z)^2}, \quad (1)$$

$$d_{n,I} = \begin{cases} \frac{S_{n,I}}{\tau}, & \text{if } \frac{S_{n,I}}{\tau} < 1, \\ 1, & \text{else,} \end{cases} \quad (2)$$

where $S_{n,I}$ denotes minimum distance from the n -th point p_n on the coronary artery to the chamber surface point set O_I . In addition, $d_{n,I}$ is the normalized distance of $S_{n,I}$, and τ is an empirical parameter, which is set as $\tau = 50\text{mm}$ in this study.

B. Hierarchical Topology Learning

The encoder of the ResUNet model within the HTL module takes advantage of multi-task learning and produces features reflecting key-point detection, centerline heat map regression, and connectivity prediction tasks. The BAI scheme involved in decoders moves a step further towards stronger feature sharing. Specifically, the proposed hierarchical topology learning (HTL) module takes the distance field maps and the close-up region-of-interest as input and consists of three branches dedicated for 1) key points (bifurcation and endpoints) detection for preserving vascular tree-topology, 2) centerline heatmap regression for preserving skeleton structure, and 3) neighbor connectivity prediction for preserving neighboring topological continuity. The feature representations of three branches are integrated by a proposed bottom-up attention interaction module (BAI). Notably, the final vessel segmentation mask arises from the neighbor connectivity prediction branch.

1) Key-Point Detection: Bifurcation points and endpoints of each coronary branch can provide strong reference for the tree-like vascular structure. Therefore, we employ a key-point detection branch to guide vessel segmentation. Inspired by [39], we use ResUNet as backbone to generate a probability map from the input image. The probability map represents the probability value of each point being predicted as the key point. We subsequently calculate the weighted Hausdorff distance (WHD) [39] between the predicted points and ground truth to estimate their similarity. Notably, the method avoids the need for a predefined maximum number of points. More details can be found in [40] and [41]. The WHD is formulated as follows:

$$\begin{aligned} \mathcal{L}_{WHD}(u_x, \Omega_{gt}) = & \frac{1}{\Omega + \epsilon} \sum_{x \in \Omega_{pd}} u_x \min_{y \in \Omega_{gt}} d(x, y) \\ & + \frac{1}{|\Omega_{gt}|} \sum_{y \in \Omega_{gt}} M_\alpha [u_x d(x, y) \\ & + (1 - u_x) d_{max}], \end{aligned} \quad (3)$$

$$M_\alpha [f(x)] = \left(\frac{1}{|\Omega_{pd}|} \sum_{x \in \Omega_{pd}} f^\alpha(x) \right)^{\frac{1}{\alpha}}, \quad (4)$$

where $u_x \in [0, 1]$ is probability value output of the network at coordinate index x . $\Omega = \sum_{x \in \Omega_{pd}} u_x$, refers to the summation of Ω_{pd} , and ϵ is set as 10^{-4} . $d(x, y)$ is the Euclidean distance between two points, where the variables x and y denote the position indices of the coordinate in the probability map Ω_{pd} and the true key point in Ω_{gt} , respectively. Moreover, $|\Omega_{gt}|$ denotes a set of true key points (the ground truth). d_{max} is the maximum Euclidean distance between possible coordinates in Ω_{pd} , i.e., the distance between the coordinates of the vertices on the diagonal points. $M_\alpha[\cdot]$ represents the generalized mean with the exponent $\alpha = -1$.

2) Centerline Heatmap Regression: Since the centerline only covers single isolated voxel on each cross-section, a little position bias of those voxels may cause huge deviation in centerline prediction, we employ 3D centerline heatmap regression instead of predicting centerline directly. We introduce a centerline heatmap unit (CHU) with a 3D adaptive

Gaussian filter G_t to generate a locally normalized centerline heatmap as shown in Fig. 2. Formally, the Gaussian filter $G_t \sim N(0, \sigma_t^2)$ is defined as below:

$$\sigma_t = \sqrt{\frac{1}{L} \sum_{l=1 \dots L} (B_t^l - C_t)^2}, \quad (5)$$

$$G_t = e^{-\frac{(B_t^l - C_t)^2}{2\sigma_t^2}}, \quad (6)$$

where C_t denotes the coordinate of the t -th point on centerline, and B_t^l denotes the l -th coordinate of the boundary points on the cross-section passing through C_t and perpendicular to the centerline. σ_t is the kernel width of G_t , which can be approximated as the vascular radius. Such a design of G_t enables the adjustment of heatmap shape based on the vascular diameter, ensuring dynamic adaptation to different vascular branches. As a result, the CHU module generates centerline heatmaps that impose skeleton consistency and facilitate the continuity of the segmented vessel.

3) Neighbor Connectivity Prediction: Since neighbor connectivity plays an extremely important role for describing the context of vessel, instead of viewing vessel segmentation as classification of individual vessel points, we take the vessel connectivity with their neighbor regions into account explicitly. This perspective aligns also with previous works [32], [42]. As shown in Fig. 2, in a 3D CCTA image, vessel connectivity is defined as the connection relationship between vessel points in a cubic region such as of size $3 \times 3 \times 3$, which is denoted as neighbor connectivity. Given a voxel of the coronary vessel label P and its all 26-nearest neighbors Q , we index neighbors Q from 1 to 26 and denote each voxel pair (P, Q_u) , as a connectivity orientation. Each orientation is encoded using a 1-channel binary label. If both P and Q_u are vascular voxels, then the pair (P, Q_u) is connected and the corresponding position P in the u -th channel is marked as 1. Otherwise, we mark it as 0 to represent disconnection. This yields a 27-channel vectors encapsulating the center point P and its 26 neighbors Q .

Specifically, in the training phase, the NCU branch generates the 27-channel prediction vectors for individual voxels, resulting in a 27-channel probability maps that quantifies connectivity likelihood. During inference, the 27-channel probability maps need to be fused into a binary map as the final segmentation result. The fusion criterion entails that pairwise voxels should be consistent with each other in connectivity. For instance, if voxel P is connected to its neighbor Q_u , then voxel P on the u -th connectivity channel is marked as 1. Meanwhile, the voxel P is also within the cubic neighborhood centered on voxel Q_u , the corresponding channel for voxel P is similarly marked as 1. Fig. 3 depicts a typical example in 2D space with 3×3 square region. The binary mask of vessel segmentation is first obtained from the probability map using a threshold of 0.5. For the center point in the yellow box, within its 3×3 neighborhood, there are 6 points connected to the vessel voxel, the fused result is then obtained by counting the amount of all the connected points. In general, the boundary points of the vessel are always connected to at least one neighboring vessel point within the 3×3 region centered on it. Following

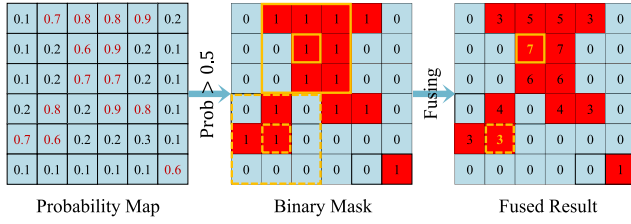


Fig. 3. Illustration of binarization of connectivity probability map and the follow-up fusion for the final fine-grained segmentation.

the pairwise fusion criterion, the fusion result at that specific center point will exceed 2. Consequently, a second threshold of 2 is used to binarize the fused values to generate the final segmented vessel. In such a way, the isolated point in the bottom right corner is effectively removed. Notably, all these operations can be performed efficiently using sliding windows and matrix operations, allowing for fast inference.

4) *Bottom-up Attention Interaction*: The above three hierarchical topology learning branches describe the vessel topological patterns from different perspectives. The connectivity branch contains a dense volume-wise representation of the vessel and generates the final segmentation result, while the centerline and key points branches are dedicated for locating skeleton and point markers of the vessel, respectively. Numerous studies [43] have shown that learning the interactions between different branches can effectively improve the overall feature representation. Here we propose a bottom-up attention interaction (BAI) module to propagate the dense-level representations (e.g., neighbor connectivity) to the sparse-level branches (e.g., centerline) to improve their performance with detailed complementary information.

To retain branch-specific features, we maintain the first and last blocks in each decoder untouched while the second and third blocks are enhanced by the BAI modules, strategically. The BAI modules are added to the second and third blocks, strategically propagating dense representations to sparse ones and enhancing overall features. The BAI module is built on cross-attention mechanism [44], as shown in Fig. 2. The ‘Query’ (Q) is from sparse feature branch such as centerline branch, denoted by F_2 , while the pair of ‘Key’ (K) and ‘Value’ (V) are from the relative dense branch such as connectivity branch, denoted as F_1 , which can be formulated as

$$F'_2 = \text{MLP} \left(\text{LN} \left(F_2 + \left[(Q(F_2) \cdot K(F_1))^T \right] * V(F_1) \right) \right) + \left(F_2 + \left[(Q(F_2) \cdot K(F_1))^T \right] * V(F_1) \right), \quad (7)$$

where F'_2 is the fused feature, and MLP and LN denote multilayer perceptron and layer normalization, respectively.

C. Loss Function

We employ different losses for the individual branch. The loss functions of the key-point detection branch and the centerline regression branch are the weighted Hausdorff distance (WHD) loss $\mathcal{L}_{WHD}$ and the mean squared error (MSE) loss $\mathcal{L}_{MSE}$, respectively. For the neighbor connectivity branch, we employ the weighted cross-entropy loss $\mathcal{L}_{WCE}$ to adjust the learning bias between the vessel and background region.

Meanwhile, we introduce the Dice similarity coefficient (DSC) loss $\mathcal{L}_{DSC}$ to ensure whole vessel segmentation performance.

In the experiment, these three branches are jointly trained and the total loss function is expressed as:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{DSC} + (1 - \lambda_1) \mathcal{L}_{WCE} + \lambda_2 (\mathcal{L}_{WHD} + \mathcal{L}_{MSE}), \quad (8)$$

where λ_1 and λ_2 are the balancing weights which are empirically set as 0.5 and 0.1 through iterative experiments. Note that the ResUnet of ADE module is optimized using only the first two items of $\mathcal{L}_{total}$, i.e., $\mathcal{L}_{DSC}$ and $\mathcal{L}_{WCE}$.

IV. EXPERIMENTS

A. Datasets

In this study, we use an in-house clinical coronary arteries dataset CTA100 and a publicly available coronary arteries dataset ASOCA [45] to evaluate our framework.

1) *CTA100*: Our in-house CCTA dataset is collected by Peking Union Medical College Hospital, where the Ethics Review Board has reviewed and approved this research. A total of 100 subjects are performed five-fold cross-validation in the experiment, with the resolution of $0.3 \times 0.3 \times 0.5 \text{ mm}^3$ and the size of $512 \times 512 \times S_z$, $S_z \in \{155, 353\}$. Three experienced radiologists participated in annotating these data by using 3D-slicer software, including heart chambers annotations (i.e., left ventricle, right ventricle, left atrium, right atrium, and the whole aorta) and coronary arteries annotations. The centerline is extracted from the entire vascular annotation using the 3D thinning algorithm [46]. Subsequently, vascular endpoints and bifurcation points are identified based on this centerline. Endpoints are defined by having only one neighboring pixel point, while bifurcation points consist of at least three neighboring pixel points. This process establishes the ground truth for key points.

2) *ASOCA*: Automated Segmentation of Coronary Arteries Dataset (ASOCA) is a public dataset in MICCAI-2020 challenge and consists of 40 subjects for training and 20 subjects for testing. Since we only have access to the annotations of 40 subjects from the training set, we take the 40 annotated subjects as the total data and evaluate it with a five-fold cross-validation. The original images are resampled to $0.4 \times 0.4 \times 0.625 \text{ mm}^3$ with the size of $512 \times 512 \times S_z$, $S_z \in \{168, 334\}$. Note that the ASOCA data are annotated only for coronary artery and not for heart chambers. The ground truth of the centerline and keypoints for each data are obtained consistently following the same procedure as the CTA100 data.

B. Evaluation Metrics & Experimental Setup

1) *Evaluation Metrics*: To quantitatively assess the effectiveness of vessel segmentation, we employ three levels of metrics, including volume-wise metrics (Dice Similarity Coefficient (DSC), True Positive Rate (TPR), Hausdorff Distance (HD), and Average Surface Distance (ASD)), as well as centerline-wise and key points-wise metrics. For the centerline-wise metrics, we incorporate the cDice score [37] to evaluate skeleton quality. Furthermore, we utilize Correctness (Corr.), Completeness (Compl.), and Quality (Qual.),

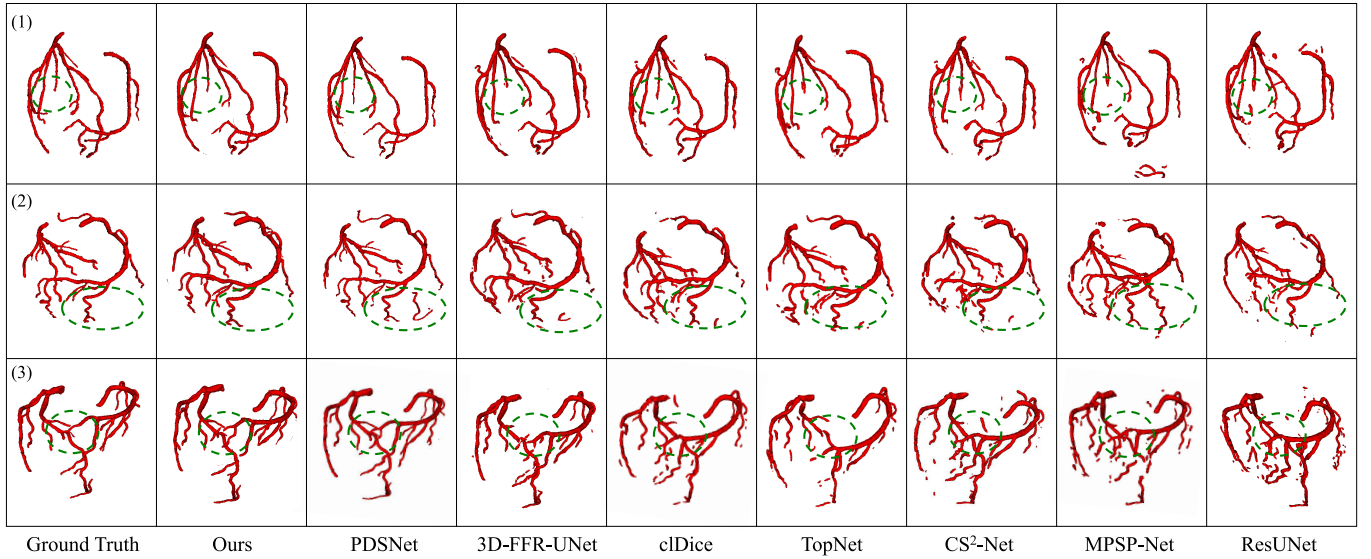


Fig. 4. Qualitative comparison for three typical cases by eight different methods. The green dashed circles highlight the regions for better visual comparison.

which reflect the overall goodness of the vascular skeleton, and are defined as:

$$Corr. = \frac{|\bar{Y}_{cl} \cap Y|}{|\bar{Y}_{cl}|}, \quad (9)$$

$$Compl. = \frac{|Y_{cl} \cap \bar{Y}|}{|Y_{cl}|}, \quad (10)$$

$$Qual. = \frac{|\bar{Y}_{cl} \cap Y| + |Y_{cl} \cap \bar{Y}|}{|\bar{Y}_{cl}| + |Y_{cl}|}, \quad (11)$$

where Y and Y_{cl} denote the ground truth of the vessel mask and centerline, respectively. $\bar{Y}$ and $\bar{Y}_{cl}$ are the corresponding predictions, respectively.

The point-wise metric we utilize is the Mean Per Joint Position Error (MPJPE), which is calculated by Euclidean distance between the predicted points and the ground truth points as follows:

$$MPJPE = \frac{1}{M} \sum_{i=1}^M \|\Omega_i - \bar{\Omega}_i\|_2, \quad (12)$$

where M is the number of key points, Ω_i and $\bar{\Omega}_i$ are respectively the ground truth and prediction of the key points.

2) Experimental Setup: We conducted quantitative and qualitative comparison with several state-of-the-art methods on both datasets using five-fold cross-validation. All experiments were implemented using Pytorch on two NVIDIA Tesla A100 GPUs. We trained the networks using Adam optimizer with an initial learning rate of 10^{-4} . The mini-batch size was set as 4 and we used 800 epochs for training. The number of blocks in BAI module was chosen as two. Moreover, we randomly cropped sub-volumes with a size of $256 \times 256 \times 64$ as the input in the training phase. In the testing phase, sliding windows of the same size were utilized with a half window overlap and cover the entire volume. Finally, the predicted results of multiple sub-volumes are combined as the final result.

C. Comparisons With State-of-the-Art Methods

To evaluate the performance of our framework, we carried out comparison with several representative deep-learning based vascular segmentation methods, including ResUNet [13], MPSPNet [28], CS²-Net [29], cLDice [37], TopNet [36], 3D-FFR-UNet [30], and PDSNet [20]. The quantitative and qualitative results are presented in Table I and in Fig. 4, respectively.

1) Quantitative Evaluation: Table I reports quantitative comparison between different methods for coronary artery segmentation in DSC, TPR, ASD, and HD, centerline in cLDice, Corr., Compl., and Qual., and position error of key points in MPJPE. Note that the proposed framework and PDSNet (our previous work) are the only two methods that incorporate centerline extraction branches. For the other methods, we employ a 3D thinning algorithm [46] to extract the centerline from the segmented vascular mask. Specifically, compared with the baseline ResUNet, our method achieves significant improvement in all the metrics, e.g., DSC increases from 70.25% to 82.14% on CTA100 and from 82.03% to 86.08% on ASOCA. In the meantime, our method surpasses PDSNet with a performance gain of 1.78% (80.36% vs. 82.14%) in DSC on CTA100 and 1.37% (84.71% vs. 86.08%) on ASOCA. Moreover, our method achieves also the best performance for centerline regression in all the three metrics. It is worth noting that all these competing methods do not consider artery-to-heart dependency and comprehensive topology of coronary artery, which usually leads to false positives and vessel discontinuities in segmentation results. Finally, paired t -tests are performed to evaluate the performance gain of our proposed method compared to the other methods, including volume metrics of DSC, TPR, 95HD, and ASD, centerline metrics of cLDice, Qual., and key point metric of MPJPE. All the p -values are lower than 0.05. The quantitative results demonstrate that our proposed method achieves state-of-the-art performance.

TABLE I
QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART METHODS IN TERMS OF VOLUME SEGMENTATION,
CENTERLINE REGRESSION, AND KEY POINTS DETECTION ON BOTH DATASETS

Dataset	Method	Volume				Centerline				Key Point
		DSC[%]↑	TPR[%]↑	ASD[mm]↓	HD[mm]↓	clDice[%]↑	Corr.[%]↑	Compl.[%]↑	Qual.[%]↑	MPJPE[mm]↓
CTA100	ResUNet [13]	70.25±1.46	76.38±1.50	1.53±0.06	9.33±0.22	80.89±2.10	78.85±3.17	83.05±1.66	80.05±1.99	-
	MPSPNet [28]	73.60±1.17	78.89±1.09	0.99±0.06	7.08±0.24	82.33±1.89	82.47±2.81	82.19±1.89	82.08±1.98	-
	CS ² -Net [29]	76.53±1.29	83.47±1.50	1.03±0.04	7.33±0.31	84.07±2.05	85.33±2.63	82.86±2.21	83.71±1.65	-
	TopNet [36]	76.65±1.02	79.98±1.14	0.86±0.08	6.35±0.34	88.35±1.83	89.13±2.06	87.60±1.63	88.05±1.43	-
	clDice [37]	76.72±0.86	84.06±1.26	0.82±0.02	6.29±0.18	87.33±1.96	86.01±2.32	88.69±1.76	86.99±1.58	-
	3D-FFR-Unet [30]	79.35±1.04	83.53±1.38	1.14±0.11	6.19±0.50	87.39±1.63	86.08±1.85	88.76±1.43	87.25±1.19	-
	PDSNet [20]	80.36±0.93	86.84±1.14	0.79±0.04	6.50±0.15	89.84±1.60	87.35±1.62	92.47±1.62	90.26±1.35	4.97±1.92
	Ours	82.14±1.15	88.03±1.37	0.80±0.07	5.68±0.23	92.30±1.48	91.14±1.39	93.48±1.82	92.59±1.17	4.26±1.76
ASOCA	ResUNet [13]	82.03±1.39	78.35±1.42	0.81±0.23	7.51±0.85	85.04±1.97	86.45±2.03	83.69±2.21	84.44±1.25	-
	MPSPNet [28]	83.08±1.46	77.58±1.50	0.93±0.37	6.91±0.62	84.28±2.24	85.00±3.33	83.57±1.95	84.29±2.13	-
	CS ² -Net [29]	82.80±1.51	81.40±1.66	1.08±0.54	7.11±0.68	84.63±2.06	86.50±2.75	82.85±1.93	84.93±1.56	-
	TopNet [36]	82.54±1.03	81.63±1.41	0.74±0.16	6.03±0.44	88.25±1.92	88.86±2.05	87.65±1.84	88.20±1.48	-
	clDice [37]	83.57±1.25	82.90±1.36	0.69±0.18	6.64±0.41	86.91±1.81	86.77±2.14	87.05±1.68	88.51±1.72	-
	3D-FFR-Unet [30]	83.85±1.30	78.49±1.49	0.78±0.14	6.77±0.52	86.47±1.86	89.06±1.94	84.02±1.85	86.10±1.39	-
	PDSNet [20]	84.71±1.46	79.62±1.37	0.42±0.21	6.12±0.57	88.29±1.69	85.94±2.08	90.78±1.57	87.69±1.45	5.04±1.65
	Ours	86.08±1.21	82.54±1.69	0.41±0.13	5.25±0.43	91.68±1.58	94.02±1.59	89.47±1.61	91.88±1.34	4.15±1.44

2) Qualitative Evaluation: Visual comparison of three typical cases are shown in Fig. 4. As can be seen that our method generates superior vascular masks than other competing methods in most cases, which coincides with the quantitative comparison. Specifically, for less complicated structures (first row), all eight methods can produce satisfactory visual performance at first glance, but the competing methods can hardly maintain the details, especially the end of the vessel (bifurcation in the green circle). For more complicated cases (last two rows), our method can well preserve the anatomy and topology of coronary artery and significantly outperforms the competing methods especially in the marked region by the green dashed line. The main reasons are that most background is eliminated by the ADE module and the false positive of background rarely appears in our results. Moreover, through hierarchical topology-preserving constraints and feature interaction, our results maintain good topology continuity in each coronary branch, unlike other methods that have a large number of discontinuities. The qualitative results further verify the effectiveness of our method for coronary artery segmentation.

D. Segmentation Performance of Heart Chamber and Dilated Vessel

We first verify the performance of the heart chamber segmentation and dilated vessel segmentation on the CTA100 dataset and we summarize the results in Table II. The mean DSC of each chamber and dilated vessel reach 95% and 87%, respectively, and the TPR reaches above 93% or more, indicating that the majority of ROIs are accurately segmented. Fig. 5

TABLE II
QUANTITATIVE SEGMENTATION PERFORMANCE OF HEART
CHAMBER AND DILATED VESSEL ON CTA100 DATASET

Organ	DSC[%]↑	TPR[%]↑	ASD[mm]↓	HD[mm]↓
Aorta	96.51±0.49	96.65±0.82	0.53±0.11	1.23±0.15
Left Atrium	96.78±0.21	96.92±0.64	0.48±0.09	1.46±0.11
Right Atrium	95.00±0.71	95.21±1.14	0.54±0.13	1.56±0.16
Left Ventricle	94.99±0.94	95.14±0.86	0.67±0.16	1.71±0.14
Right Ventricle	95.74±0.95	97.21±0.75	0.56±0.15	1.58±0.08
Dilated Vessel	87.36±1.14	92.86±1.52	0.72±0.37	5.34±0.58

demonstrates the segmented chambers and dilated vessel. It can be seen that the predicted results are highly consistent with the ground truth both in 2D and 3D illustrations. The last column shows the segmentation of dilated vessel, where the pink mask is the prediction and the red mask depicts the undilated coronary artery (ground truth). It can be seen that the predicted mask contains most of the ground truth regions, providing a reliable basis for the estimation of the distance field map.

E. Ablation Study

In the following experiments, for brevity, we denote the coarse segmentation (only by ADE module) as bNet-a, the integration of key-point detection as bNet-a-p, the use of centerline regression as bNet-a-p-l, the use of neighbor

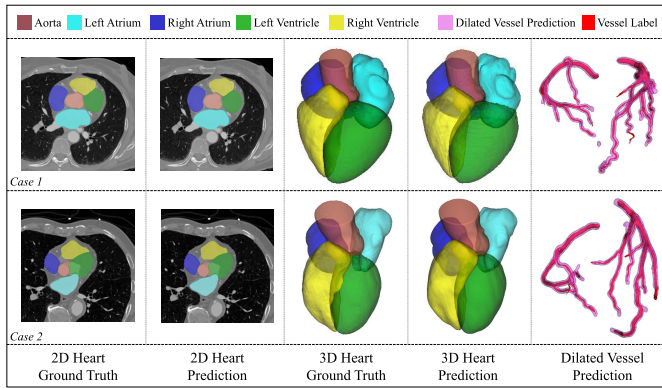


Fig. 5. Segmentation results of heart chambers and dilated vessel for two typical cases on CTA dataset.

TABLE III

EVALUATING THE IMPACT OF CHAMBER SEGMENTATION QUALITY ON VESSEL SEGMENTATION PERFORMANCE ON CTA100 DATASET

Method	DSC[%]↑	TPR[%]↑	ASD[mm]↓	HD[mm]↓
bNet	70.25±1.46	76.38±1.50	1.53±0.06	9.33±0.22
bNet-a(85)	68.54±1.74	73.19±1.66	1.89±0.14	11.46±0.53
bNet-a(90)	71.06±1.62	75.97±1.30	1.60±0.11	9.12±0.36
bNet-a(95)	73.26±1.25	78.67±1.29	1.19±0.06	8.54±0.28
bNet-a(100)	73.11±1.16	78.95±1.34	1.31±0.10	8.37±0.25

connectivity prediction as bNet-a-p-l-n and the full capacity with BAI modules as FullNet.

1) *Impact of Heart Chamber Segmentation*: To further investigate the impact of heart chamber segmentation on coarse segmentation performance, we conducted additional experiments generating five different levels of accuracy for heart segmentation by 1) Using baseline ResUNet (bNet) to segment coronary artery from raw images; 2) Adjusting the ResUNet parameters to achieve a Dice score of 85% for heart chamber segmentation, and replacing the heart mask required in ADE module, denoted as bNet-a(85); 3) Similar to configuration 2, using the ResUNet with the chamber segmentation of 90% in Dice score, denoted as bNet-a(90); 4) Using the ResUNet with the chamber segmentation of 95% in Dice score (proposed), denoted as bNet-a(95); 5) Using the ground truth of heart chambers in the ADE module, denoted as bNet-a(100).

Table III shows that the performance of coarse-grained vessel segmentation gradually improves as the accuracy of the added chamber masks increases. When accuracy of chamber masks reaches a certain level, segmentation performance plateaus. We can see that the variant bNet-a (DSC: 73.26%) actually generates the best performance, which indicates that higher accuracy of chamber masks might be not necessary. Hence, we use bNet-a in the following experiments. Moreover, low-precision chamber masks may interfere with the network learning and result in performance degradation, as shown for the example comparing bNet-a(85) with bNet-a(95).

2) *Effectiveness of HTL and BAI Modules*: In Table IV, we summarize the segmentation performance of the eight model variants, *i.e.*, bNet-a, bNet-p, bNet-p-l, bNet-p-l-n,

TABLE IV

QUANTITATIVE RESULTS FOR ABLATION STUDY OF DIFFERENT MODEL VARIANTS

Method	Dataset	DSC[%]↑	TPR[%]↑	ASD[mm]↓	HD[mm]↓
bNet-p	CTA100	72.31±1.37	77.52±1.34	1.30±0.08	9.01±0.32
bNet-p-l	CTA100	76.02±1.21	82.96±1.35	1.12±0.07	7.03±0.27
bNet-p-l-n	CTA100	79.86±1.02	85.62±1.26	0.90±0.05	6.15±0.20
bNet-a	CTA100	73.26±1.25	78.67±1.29	1.19±0.06	8.54±0.28
bNet-a-p	CTA100	75.19±1.33	82.03±1.29	0.98±0.05	6.22±0.35
bNet-a-p-l	CTA100	78.47±1.12	84.17±1.24	1.00±0.06	6.31±0.23
bNet-a-p-l-n	CTA100	81.03±0.89	86.92±1.21	0.71±0.04	5.38±0.18
FullNet	CTA100	82.14±1.15	88.03±1.37	0.80±0.07	5.68±0.23
FullNet	CTA100ASOCA	77.14±1.24	91.14±1.48	2.01±0.51	9.75±1.52
FullNet	ASOCA	86.08±1.21	82.54±1.69	0.41±0.13	5.25±0.43
FullNet	ASOCACTA100	73.63±2.04	68.96±2.48	1.39±0.85	11.87±2.17

bNet-a-p, bNet-a-p-l, bNet-a-p-l-n, and FullNet. Comparing the performance of bNet-p, bNet-p-l, and bNet-p-l-n, a gradual improvement can be observed under all the evaluation metrics. This trend indicates that the introduced hierarchical topology constraints contribute positively to the segmentation results. Furthermore, this positive trend continues to advance with the incorporation of the ADE module, as indicated by the comparison results among bNet-a-p, bNet-a-p-l, and bNet-a-p-l-n in Table IV. When comparing bNet-a-p-l-n with bNet-a, we can see that the use of hierarchical topology learning improves the performance of DSC, TPR, ASD, and HD from 73.26%, 78.67%, 1.19 mm, and 8.54mm to 81.03%, 86.92%, 0.71mm, and 5.38mm, respectively, which validates the effectiveness of the proposed HTL model. Besides, when comparing bNet-a-p-l-n and FullNet, we can observe the effectiveness of feature interaction among the three branches by the proposed BAI.

To further explore the impact of different number of Transformer blocks within the BAI module on segmentation performance, we conducted ablation study and show the results in Table V. Comparing FullNet's performance using 0, 1, and 2 Transformer blocks, we observe a consistent increase in DSC from 81.03% to 82.14%. However, when continue increasing Transformer blocks from 2 to 4 blocks, the DSC drops from 82.14% to 81.20%. It indicates that although integrating Transformer blocks enhances the BAI module, beyond a certain level of complexity, the benefits plateau or diminish. In experiments, we opt for 2 Transformer blocks, striking a balance between enriching features and maintaining computational complexity.

To further demonstrate the merit of our method, we show the surface-to-surface distance and compare with the ground truth in Fig. 6. It is shown that our predictions exhibit remarkable proximity to the ground truth in both 2D and 3D visualizations. The 3D visualization reveals the consistency with the ground truth's spatial distribution. Notably, our method reduces the occurrence of over-segmented or broken vessel branches, providing additional evidence for the reliability of our method.

TABLE V
QUANTITATIVE RESULTS USING DIFFERENT NUMBERS OF
TRANSFORMER BLOCKS FOR THE BAI MODULE
ON CTA100 DATASET

Method	Number	DSC[%]↑	TPR[%]↑	ASD[mm]↓	HD[mm]↓
FullNet	0	81.03±0.89	86.92±1.21	0.71±0.04	5.38±0.18
FullNet	1	81.37±1.17	87.15±1.33	0.86±0.09	5.74±0.25
FullNet	2	82.14±1.15	88.03±1.37	0.80±0.07	5.68±0.23
FullNet	3	82.11±1.06	88.09±1.39	0.79±0.07	5.69±0.22
FullNet	4	81.20±1.12	87.06±1.34	0.83±0.08	5.72±0.24

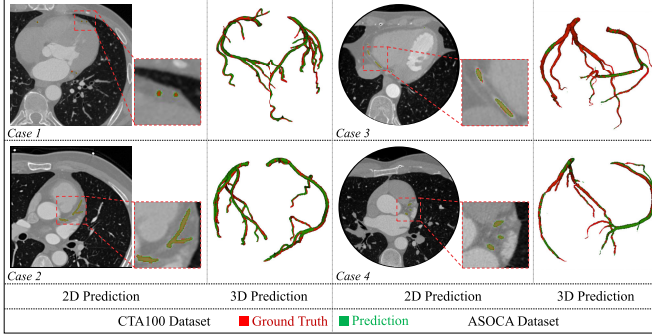


Fig. 6. Visualization of surface-to-surface distance between ground truth and prediction.

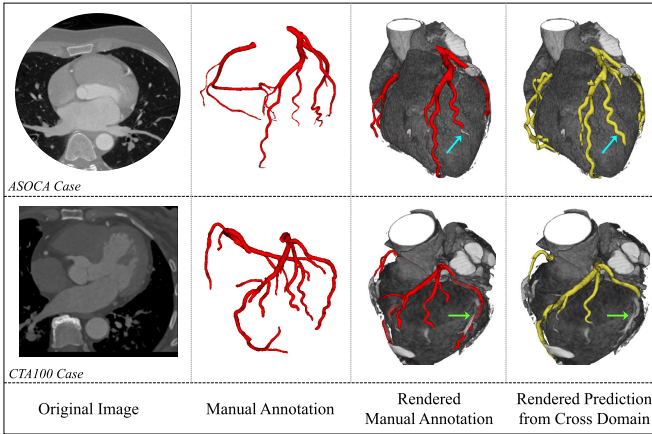


Fig. 7. The segmentation results of cross-domain evaluation. The 1st column shows the results of testing ASOCA data using the CTA100 trained network, which appears to be over-segmented compared to the manual annotation but remains consistent with actual vessel gold standard (cyan arrow). The 2nd column shows the opposite scenario, depicting the under-segmentation case of testing CTA100 data (green arrow).

3) Effectiveness of Cross-Domain Segmentation: To verify the robustness and generalization of our framework, we perform cross-domain segmentation on two datasets. Specifically, CTA100 data was taken as the training set and ASOCA as the testing one, denoted as CTA100ASOCA. Conversely, ASOCA data was taken as the training set and CTA100 for testing, denoted as ASOCACTA100. As shown in Table IV, compared with the results of training and testing within the same dataset (i.e., 5th row and 6th row, 7th row and 8th row), DSC score of CTA100ASOCA and ASOCACTA100 decrease by 5.00% (82.14% vs. 77.14%) and 12.45% (86.08% vs. 73.63%),

respectively. We find that CTA and ASOCA data have significant differences in the image domain due to different data sources and scanners as shown in Fig. 7. Additionally, ASOCA data only contains annotation for the major vascular branch, whereas CTA100 undergoes exhaustive labeling. As a result, models trained on CTA100 tend to over-segment when applied to ASOCA during testing, resulting in a high TPR (91.14%). Conversely, models trained on ASOCA tend to under-segment when testing on CTA100, as shown in the last column of Fig. 7.

V. DISCUSSION AND CONCLUSION

In this paper, we propose an anatomy- and topology-preserving framework for accurate coronary artery segmentation. This framework consists of an anatomical dependency encoding module and a hierarchical topology learning module. The former module performs coarse segmentation and is based on which embeds the anatomical dependency between coronary artery and heart chambers in distance field maps and crops the zoom-up region-of-interest. The latter one refines the segmentation by resorting to hierarchical topology-preserving constraints including key points, centerlines, and neighbor connectivity. These multi-level representations are interacted and integrated by a proposed bottom-up attention interaction module. Extensive experiments demonstrate that the proposed framework outperforms the state-of-the-art methods significantly and achieves promising performance in average DSC scores of 86.08% and 82.14% on public dataset ASOCA and in-house dataset CTA100, respectively.

Despite the above innovations and advantages, our proposed method still has some limitations in cross-domain segmentation, including: 1) The quality of topological representations could be improved by incorporating more medical expert knowledge to enhance coronary artery segmentation performance; and 2) Potential data discrepancies arising from labeling differences across various centers could be avoided by developing label noise and non-exhaustive labeling methods. We intend to address these limitations in our future work. These initiatives offer insights and guidance for advancing coronary artery segmentation performance across various clinical scenarios.

REFERENCES

- [1] C. Collet et al., "Left main coronary artery disease: Pathophysiology, diagnosis, and treatment," *Nature Rev. Cardiol.*, vol. 15, no. 6, pp. 321–331, Jun. 2018.
- [2] S. S. Virani et al., "Heart disease and stroke statistics-2021 update: A report from the American Heart Association," *Circulation*, vol. 143, no. 8, pp. e254–e743, 2021.
- [3] R. Marano et al., "CCTA in the diagnosis of coronary artery disease," *La Radiol. Medica*, vol. 125, no. 11, pp. 1102–1113, Nov. 2020.
- [4] P. W. Serruys et al., "Coronary computed tomographic angiography for complete assessment of coronary artery disease: JACC state-of-the-art review," *J. Amer. College Cardiol.*, vol. 78, no. 7, pp. 713–736, 2021.
- [5] D. Cao et al., "Non-cardiac surgery in patients with coronary artery disease: Risk evaluation and periprocedural management," *Nature Rev. Cardiol.*, vol. 18, no. 1, pp. 37–57, Jan. 2021.
- [6] M. Zreik et al., "Deep learning analysis of coronary arteries in cardiac CT angiography for detection of patients requiring invasive coronary angiography," *IEEE Trans. Med. Imag.*, vol. 39, no. 5, pp. 1545–1557, May 2020.

- [7] X. Zhang, Z. Cui, J. Feng, Y. Song, D. Wu, and D. Shen, "CorLab-Net: Anatomical dependency-aware point-cloud learning for automatic labeling of coronary arteries," in *Proc. Int. Workshop Mach. Learn. Med. Imag.*, 2021, pp. 576–585.
- [8] J. M. Wolterink, R. W. van Hamersvelt, M. A. Viergever, T. Leiner, and I. Išgum, "Coronary artery centerline extraction in cardiac CT angiography using a CNN-based orientation classifier," *Med. Image Anal.*, vol. 51, pp. 46–60, Jan. 2019.
- [9] A. F. Frangi, W. J. Niessen, K. L. Vincken, and M. A. Viergever, "Multiscale vessel enhancement filtering," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 1998, pp. 130–137.
- [10] S. Wang, B. Li, and S. Zhou, "A segmentation method of coronary angiograms based on multi-scale filtering and region-growing," in *Proc. Int. Conf. Biomed. Eng. Biotechnol.*, May 2012, pp. 678–681.
- [11] G. Yang et al., "Automatic centerline extraction of coronary arteries in coronary computed tomographic angiography," *Int. J. Cardiovascular Imag.*, vol. 28, no. 4, pp. 921–933, Apr. 2012.
- [12] Y. Qi et al., "Examinee-examiner network: Weakly supervised accurate coronary lumen segmentation using centerline constraint," *IEEE Trans. Image Process.*, vol. 30, pp. 9429–9441, 2021.
- [13] E. Kerfoot, J. Clough, I. Oksuz, J. Lee, A. P. King, and J. A. Schnabel, "Left-ventricle quantification using residual U-Net," in *Proc. Int. Workshop Stat. Atlases Comput. Models Heart*, 2018, pp. 371–380.
- [14] R. Gharleghi, N. Chen, A. Sowmya, and S. Beier, "Towards automated coronary artery segmentation: A systematic review," *Comput. Methods Programs Biomed.*, vol. 225, Oct. 2022, Art. no. 107015.
- [15] D. Hao et al., "Sequential vessel segmentation via deep channel attention network," *Neural Netw.*, vol. 128, pp. 172–187, Aug. 2020.
- [16] H. Zhang, D. Zhang, Z. Gao, and H. Zhang, "Joint segmentation and quantification of main coronary vessels using dual-branch multi-scale attention network," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2021, pp. 369–378.
- [17] Y. Ning, S. Zhang, X. Xi, J. Guo, P. Liu, and C. Zhang, "CAC-EMVT: Efficient coronary artery calcium segmentation with multi-scale vision transformers," in *Proc. IEEE Int. Conf. Bioinf. Biomed. (BIBM)*, Dec. 2021, pp. 1462–1467.
- [18] Z. Guo et al., "DeepCenterline: A multi-task fully convolutional network for centerline extraction," in *Proc. Int. Conf. Inf. Process. Med. Imag.*, 2019, pp. 441–453.
- [19] Y. Wang et al., "Deep distance transform for tubular structure segmentation in CT scans," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3832–3841.
- [20] X. Zhang et al., "Progressive deep segmentation of coronary artery via hierarchical topology learning," in *Proc. Int. Conf. Med. Image Comput. Comput. Assist. Intervent.*, 2022, pp. 391–400.
- [21] J. Stühmer, P. Schröder, and D. Cremers, "Tree shape priors with connectivity constraints using convex relaxation on general graphs," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 2336–2343.
- [22] S. Çimen, A. Gooya, M. Grass, and A. F. Frangi, "Reconstruction of coronary arteries from X-ray angiography: A review," *Med. Image Anal.*, vol. 32, pp. 46–68, Aug. 2016.
- [23] S. Bohlender, I. Oksuz, and A. Mukhopadhyay, "A survey on shape-constraint deep learning for medical image segmentation," *IEEE Rev. Biomed. Eng.*, vol. 16, pp. 225–240, 2023.
- [24] S. Minaee, Y. Boykov, F. Porikli, A. Plaza, N. Kehtarnavaz, and D. Terzopoulos, "Image segmentation using deep learning: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 7, pp. 3523–3542, Jul. 2022.
- [25] Y. Yuan, L. Zhang, L. Wang, and H. Huang, "Multi-level attention network for retinal vessel segmentation," *IEEE J. Biomed. Health Informat.*, vol. 26, no. 1, pp. 312–323, Jan. 2022.
- [26] S. Li et al., "Automatic dental plaque segmentation based on local-to-global features fused self-attention network," *IEEE J. Biomed. Health Informat.*, vol. 26, no. 5, pp. 2240–2251, May 2022.
- [27] Y. Li, J. Ni, A. Elazab, and J. Wu, "Multiple self-attention network for intracranial vessel segmentation," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2021, pp. 1–8.
- [28] X. Zhu, Z. Cheng, S. Wang, X. Chen, and G. Lu, "Coronary angiography image segmentation based on PSPNet," *Comput. Methods Programs Biomed.*, vol. 200, Mar. 2021, Art. no. 105897.
- [29] L. Mou et al., "CS²-Net: Deep learning segmentation of curvilinear structures in medical imaging," *Med. Image Anal.*, vol. 67, Jan. 2021, Art. no. 101874.
- [30] A. Song et al., "Automatic coronary artery segmentation of CCTA images with an efficient feature-fusion-and-rectification 3D-UNet," *IEEE J. Biomed. Health Informat.*, vol. 26, no. 8, pp. 4044–4055, Aug. 2022.
- [31] P. K. Saha, R. Strand, and G. Borgefors, "Digital topology and geometry in medical imaging: A survey," *IEEE Trans. Med. Imag.*, vol. 34, no. 9, pp. 1940–1964, Sep. 2015.
- [32] M. Kampffmeyer, N. Dong, X. Liang, Y. Zhang, and E. P. Xing, "ConnNet: A long-range relation-aware pixel-connectivity network for salient segmentation," *IEEE Trans. Image Process.*, vol. 28, no. 5, pp. 2518–2529, May 2019.
- [33] S. Leventhal, A. Gyulassy, M. Heimann, and V. Pascucci, "Exploring classification of topological priors with machine learning for feature extraction," *IEEE Trans. Vis. Comput. Graphics*, early access, Feb. 24, 2023, doi: 10.1109/TVCG.2023.3248632.
- [34] B. Kong et al., "Learning tree-structured representation for 3D coronary artery segmentation," *Computerized Med. Imag. Graph.*, vol. 80, Mar. 2020, Art. no. 101688.
- [35] X. Li, R. Bala, and V. Monga, "Robust deep 3D blood vessel segmentation using structural priors," *IEEE Trans. Image Process.*, vol. 31, pp. 1271–1284, 2022.
- [36] D. Keshwani, Y. Kitamura, S. Ihara, S. Iizuka, and E. Simo-Serra, "Top-Net: Topology preserving metric learning for vessel tree reconstruction and labelling," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2020, pp. 14–23.
- [37] S. Shit et al., "CIDice—A novel topology-preserving loss function for tubular structure segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 16555–16564.
- [38] Y. Zhang and A. C. Chung, "Deep supervision with additional labels for retinal vessel segmentation task," in *Proc. Int. Conf. Med. Image Comput. Comput. Assist. Intervent.*, 2018, pp. 83–91.
- [39] J. Ribera, D. Güera, Y. Chen, and E. J. Delp, "Locating objects without bounding boxes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6472–6482.
- [40] T. Ma, A. Gupta, and M. R. Sabuncu, "Volumetric landmark detection with a multi-scale shift equivariant neural network," in *Proc. IEEE 17th Int. Symp. Biomed. Imag. (ISBI)*, Apr. 2020, pp. 981–985.
- [41] C. Payer, D. Štern, H. Bischof, and M. Urschler, "Integrating spatial configuration into heatmap regression based CNNs for landmark localization," *Med. Image Anal.*, vol. 54, pp. 207–219, May 2019.
- [42] Y. Qin et al., "AirwayNet: A voxel-connectivity aware approach for accurate airway segmentation using convolutional neural networks," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2019, pp. 212–220.
- [43] T. Han et al., "Recursive centerline- and direction-aware joint learning network with ensemble strategy for vessel segmentation in X-ray angiography images," *Comput. Methods Programs Biomed.*, vol. 220, Jun. 2022, Art. no. 106787.
- [44] A. Vaswani et al., "Attention is all you need," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2017, vol. 30, no. 11, pp. 6000–6010.
- [45] R. Gharleghi et al., "Automated segmentation of normal and diseased coronary arteries—The ASOCA challenge," *Comput. Med. Imag. Graph.*, vol. 97, Apr. 2022, Art. no. 102049.
- [46] T. C. Lee, R. L. Kashyap, and C. N. Chu, "Building skeleton models via 3-D medial surface axis thinning algorithms," *CVGIP, Graph. Models Image Process.*, vol. 56, no. 6, pp. 462–478, Nov. 1994.